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* STep By Step process in EDA And Feature Engineering in Data science project. The main aim to convert the Raw Data → useful Data.

1. Step 1 EDA {Exploratory Data Analysis}

Raw Data :

(i) Numerical feature may be there {Histrogram, pdf}

(ii) Categorical feature.

(iii) Missing value {visualize all the graph}

(iv) Outliers {Box plot}

(v) Cleaning.

2. Step 2 Handling the Missing value {FE playlist}

3. Step 3 Handling Imbalenced dataset

4. Step 4 Treating the outliers

5. Scaling down the data {Standardzition, MinMaxscaler}

6. Step 6 Converting the categorical feature into numerical featur

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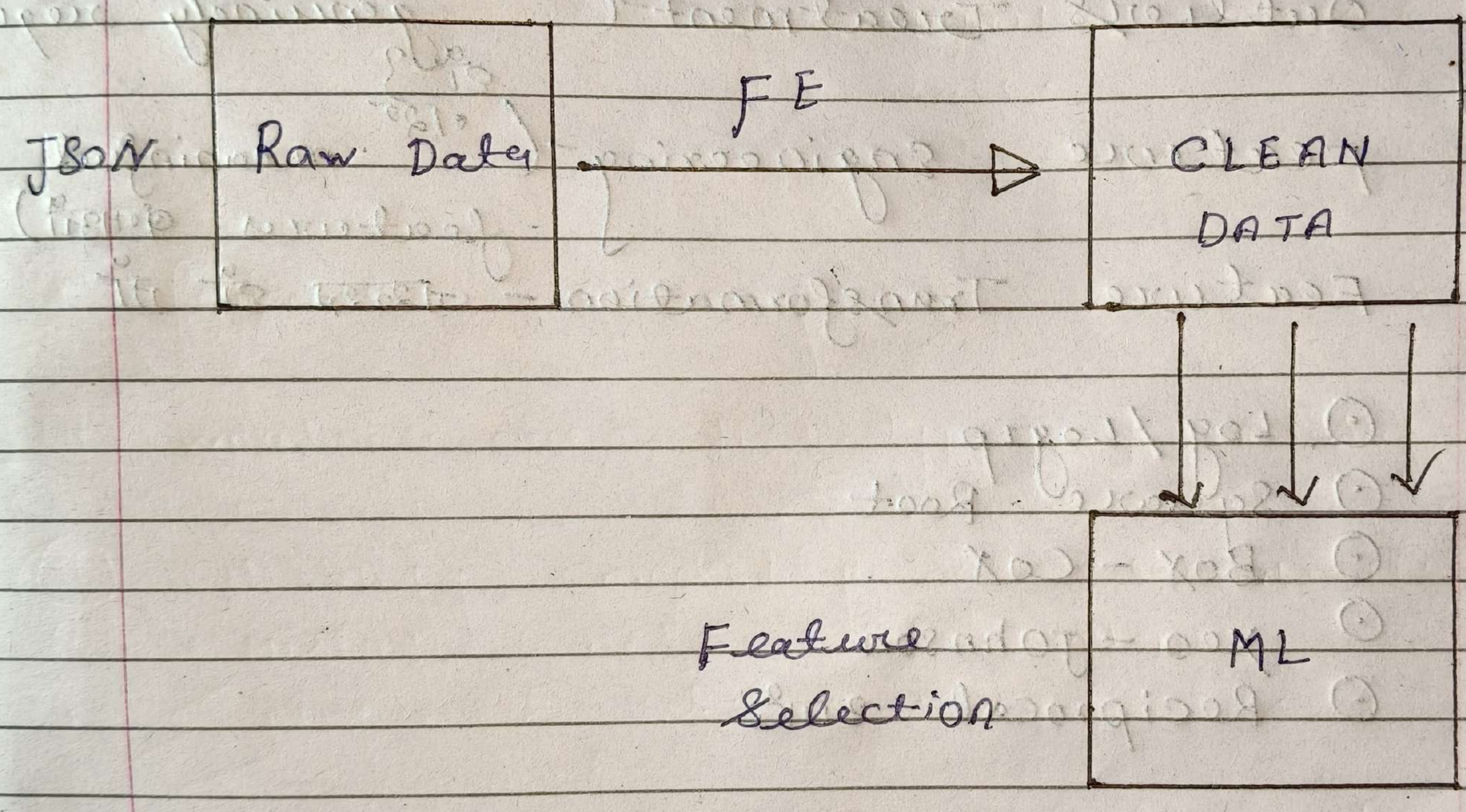
## (7) Feature Selection

(i) Correlation
(ii) K Neighbour
(iii) Chi square
(iv) Genetic Algorithms
(v) Feature Importance { Extra tree classifier }



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Chatgpt

* EDA {Exploratory Data Analysis}

1. EDA
2. Train - Test - Split
3. Missing value Handling
4. Outliers Treatment - (अगर genuinely required हो)
5. Feature Engineering - (नए meaningful features बनाओ)
6. Feature Transformation - जरूरत हो तो
    - Log / Log1p
    - Square Root
    - Box - Cox
    - Yeo - Johnson
    - Reciprocal आदि
7. Encoding
8. Feature Scaling - जरूरत हो तो
    - Standardization
    - Min / Max / Normalization
    - Robust Scaling

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(9) Imbalanced Data Handling – Classification में जरूरत होती

(10) Feature Selection

(11) Model Building

(12) Cross-validation + Evaluation

(13) Hyperparameter Tuning

(14) Final Model → Explainability → Deployment

# 2. Types of ENCODING

(I) Nominal Encoding ( Nominal categorical variable)
(II) ORDINAL Encoding ( ORDINAL categorical variable)

→ Gender State
male Delhi
Female mumbai

→ RANKING
Eduction PHD - 1
BE B.com - 2
B.Com BE - 3
PHD

(I) Nominal Encoding
- → One hot Encoding
- → one hot Encoding with many categorical
- → mean Encoding

(II) ORDINAL Encoding
- → label Encoding
- → Target guided ordinal Encoding.

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# (1) One hot Encoding

Country — Dummy variable Trap

| Country | India | Germany | France | ~~SPAIN~~ |
|---|---|---|---|---|
| India | 1 | 0 | 0 | ~~0~~ |
| Germany | 0 | 1 | 0 | ~~0~~ |
| France | 0 | 0 | 1 | ~~0~~ |
| SPAIN | | | | ~~1~~ |

n = 100

n = 99 colmns (n-1)

ADv

(1) easy to use

(11) easy to understand

DADv

(1) Break the dimentionlity

# (11) label Encoding (ORDINAL)

| Education | |
|---|---|
| BE | 1 |
| MAS | 3 |
| PHD | 4 |
| B.Com | 2 |

# (111) One hot encoding with multipal categories

KDD orange

f1

feature 50 - columns.

10 unique -> created 9 columns (n-1)

(iv) Target guided ordinal categories

| | $f_1$ | o/p | | |
|---|---|---|---|---|
| 4 → | A | 1 | → mean | 0.73 |
| 3 → | B | 1 | | 0.6 |
| 2 → | C | 0 | | 0.4 |
| 1 → | D | 1 | | 0.2 |
| | A | 0 | | 0.3 |
| | B | 0 | | 0.1 |

based on the mean value rearrange the labels and assigned rank.

(v) Mean Encoding

| $f_1$ pincode | o/p | mean |
|---|---|---|
| 843319 | 1 | 0.73 |
| 560012 | 0 | 0.5 |
| 560011 | 1 | |

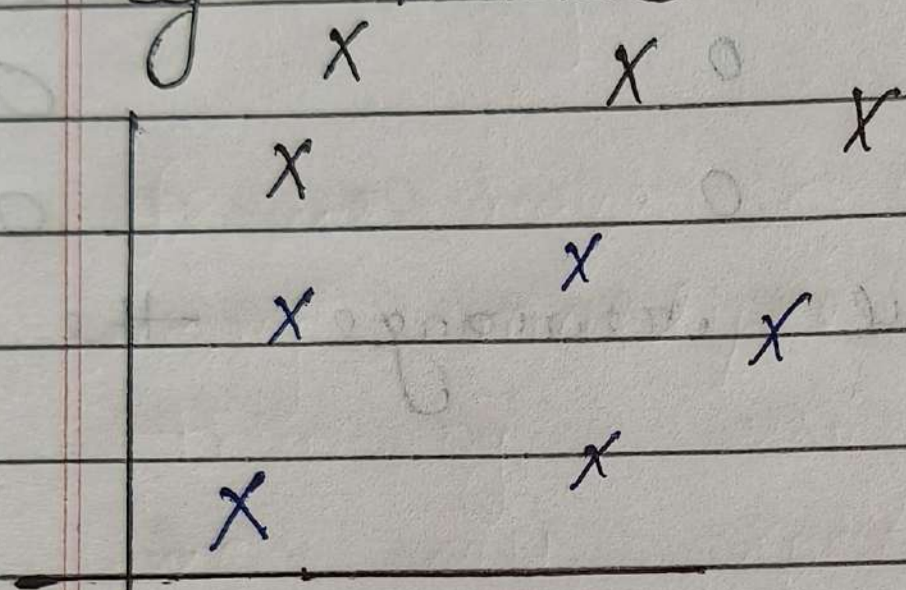
replace pincode with mean value on the basics of these perform encoding.

## * Why Do we Need to perform Feature Scaling ??

| | Height (cm) | Weight (kg) | BMI |
|---|---|---|---|
| | 180 | 78 | |
| | 170 | 84 | |

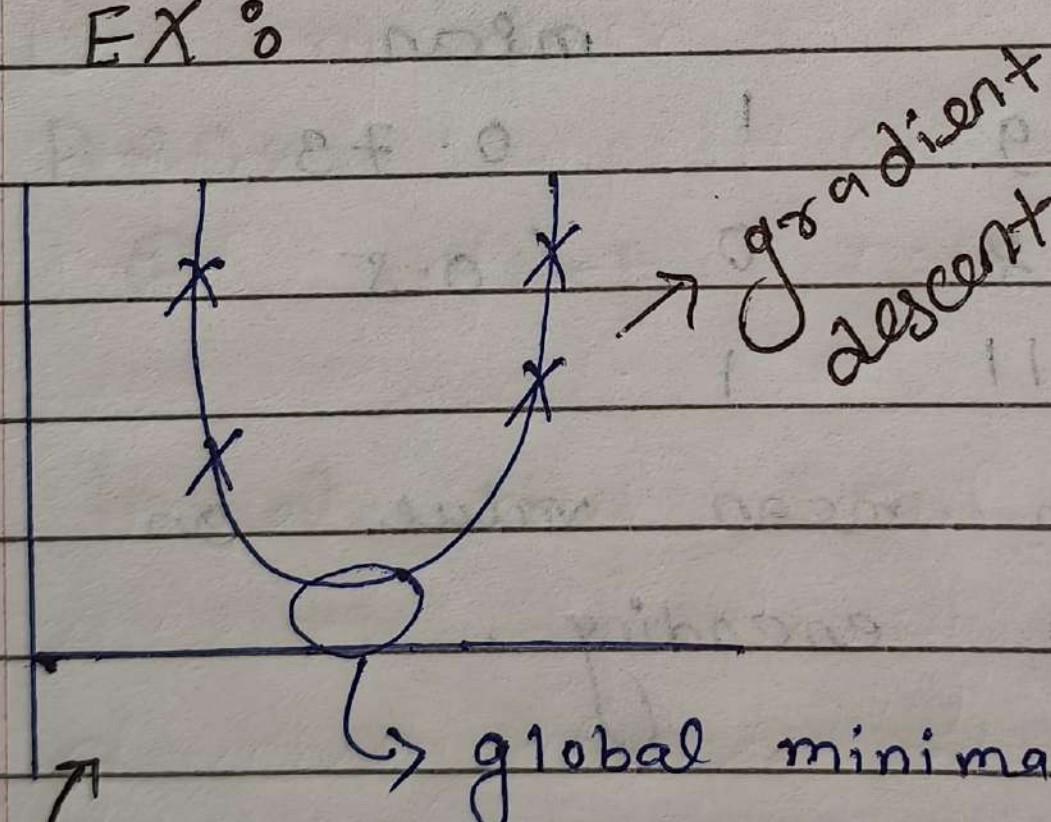
feature
- → magnitudes
- → units

eg: K- Nearest Neighbors (K-NN)



work on Eucliden distance

EX:




1. linear Regression
2. K - means
3. KNN

Note:- In all these technique you have don't use feature scaling

1. Decision Tree
2. RF
3. Xgboost

In these technique not impactful to use feature Scaling.

## * HANDLE MISSING VALUES IN CATEGORICAL VAR.

1. Delete the rows
2. Replace with the most FREQUENT VALUES
3. Apply classifier algorithm To predict
4. Apply Unsupervised ML

Train

| f1 | f2 | f3 | o/p |
| --- | --- | --- | --- |
| MALE | 23 | 24 | yes |
| ____ | 24 | 25 | NO |
| FEMALE | 25 | 26 | yes |
| MALE | 26 | 27 | yes |